

Demo Code Instruction

Severe Floods and Opioid Overdose Mortality in Appalachian Counties

Overview

This repository contains the code and data to replicate the analyses, tables, and figures in the paper. The analysis estimates the effect of severe flooding on county-level opioid overdose mortality in rural Appalachian counties, using a Sun and Abraham (2021) event-study design. **Stata** runs the regressions and exports coefficient tables; **R** reads those tables to build the figures.

Data availability

The county-by-year opioid overdose death counts derive from CDC/NCHS Restricted-Use Multiple Cause of Death micro-data and cannot be redistributed. In `floodopioid_demo.dta`, all 33 `death_*` variables are set to missing (variable names and labels are retained). Obtain the suppressed public version from [CDC WONDER](#) (counts < 10 suppressed), or the restricted micro-data via NCHS, and merge by `county_fips × year`. See `DataSources.pdf` (ICD-10 codes) and the repository `README.md`.

Software requirements

- **Stata 19** — packages `eventstudyinteract`, `estout`, `reghdfe`, `ftools` (see `InstallationGuide.pdf`).
- **R (4.3+)** — packages `ggplot2`, `dplyr`, `readr`, `patchwork`, `stringr`, `sf`, `tmap`, `tmaptools`, `haven`.

Quick start

- **1. Set one path.** In `A_Demo/democode/master.do`, set `global root` to the location of this folder on your machine.
- **2. Run Stata:** `do master.do` — runs the five analysis scripts in order and exports coefficient tables to `A_Demo/democode/output/tables/`.
- **3. Run R:** set `root` (a commented template line is provided) in each script under `A_Demo/democode/figures/`, then source them to build the figures in `A_Demo/democode/output/figures/`.

No machine-specific paths are hard-coded — you set the root path once in Stata and once in R.

Repository structure

```
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├─ README.md                start here (data + this guide)
├─ DataSources.pdf          data sources, ICD-10 codes
├─ InstallationGuide.pdf    Stata + R package setup
├─ LICENSE.pdf
├─ A_Demo/
│   └─ democode/
│       ├── master.do        Stata entry point (set root here)
│       ├── 01_main_analysis.do  main event-study regressions
│       ├── 02_subgroup_comparison.do  subgroup heterogeneity tests
│       ├── 03_robustness_checks.do  NOAA / drug / urban / drop AL,MS
│       ├── 04_migration_robustness.do  migration / composition controls
│       ├── 05_mortality_burden.do  IHS(count) burden + delta CIs
│       ├── figures/          R scripts (headers list figures)
│       └─ output/tables, output/figures
│   └─ demodata/
│       ├── floodopioid_demo.dta    analysis data (death_* suppressed)
│       ├── floodopioid_demo_codebook.pdf
│       └─ shapefiles/            state incl.; county to be downloaded
└─ B_Build/                    data-construction .do files (reference)
```

Each R script in democode/figures/ documents in its header which figures it builds, so there is no separate figure-to-code table to maintain.

Notes

- **County shapefile not included** (exceeds the host's file-size limit). Download the U.S. Census TIGER/Line county shapefile and place it in A_Demo/demodata/shapefiles/ — see README_county_shapefile.txt there.
- **County-level burden files** (burden_ihsc_*.dta) are not shipped because they derive from the restricted death data; they are regenerated by 05_mortality_burden.do.
- **B_Build/** contains the data-construction .do files that built floodopioid_demo.dta from raw FEMA, NOAA, Census, and CDC sources; included for reference (they require the restricted death data and external downloads).